

- 23** The alkene 2,4-dimethylpenta-1,3-diene reacts with two moles of HBr to give **X** as the major product.

What is the structure of **X**?

- A**  $\text{CH}_2\text{BrCH}(\text{CH}_3)\text{CH}_2\text{CBr}(\text{CH}_3)_2$
- B**  $\text{CH}_2\text{BrCH}(\text{CH}_3)\text{CHBrCH}(\text{CH}_3)_2$
- C**  $(\text{CH}_3)_2\text{CBrCHBrCH}(\text{CH}_3)_2$
- D**  $(\text{CH}_3)_2\text{CBrCH}_2\text{CBr}(\text{CH}_3)_2$

- 24** Which compound can form an organic product with molecular formula  $\text{C}_5\text{H}_8\text{O}_2$  when heated